Ayush Kansal

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EDUCATION

University of Illinois Urbana-Champaign — *Bachelor of Science in Computer Science* Expected May 2029
- Relevant Coursework: Discrete Structures (CS 173), Linear Algebra (MATH 257), Calculus III (MATH 241).

University of California, Berkeley — *concurrent coursework* Jun 2024 – Aug 2025
- CS 61A Structure and Interpretation of Computer Programs (A–); CS 61BL Data Structures (A); Data C8 Foundations of Data Science (A).

TECHNICAL SKILLS

- **Languages:** Python, Java, C, Swift, Kotlin, JavaScript, TypeScript
- **Computer Science Fundamentals:** data structures, algorithms, object-oriented design, discrete math, linear algebra
- **Systems & Tools:** Linux/Unix, Git, Bazel, FreeRTOS, CAN bus, Xcode
- **ML & AI:** PyTorch, NumPy, LLM integration (Claude API, Apple Foundation Models), computer vision, speech recognition

PROJECTS

- **Lotus Bud Academy Daycare Platform** (lotusbudacademy.com/software) — sole developer of a full-stack iOS, Android, and web platform in production: daily reports, photos, attendance, messaging, invoicing, and an AI assistant.
- **Edge ML Inference Benchmark** (github.com/AtomicAyush/edge-ml-bench; Core ML, ONNX Runtime, Swift) — benchmarked YOLOv8n at FP32/FP16/INT8 on CPU, GPU, and Neural Engine (Neural Engine FP16 8× faster than ONNX Runtime CPU FP32, equal mAP); fixed INT8's 0 mAP and closed 78% of its accuracy gap (per-channel calibration, mixed precision); iPhone: 1.6 ms image-to-boxes.
- **API Autopilot** (Node.js, TypeScript compiler API; co-developed with Ishaan Manoor) — open-source CLI and GitHub Action that rewrites JS/TS code for breaking SDK releases; extended its call-site tracing from 16% to 98% of Mux SDK calls across 32 public repos; on Cloudflare's SDK 6→7 its drafted rules fixed 125 of 246 upgrade type errors, introducing none.
- **NoteFlash** (iOS/iPadOS; SwiftUI, SwiftData, Foundation Models) — flashcard app that turns notes, scanned pages, PDFs, slides, and Google Drive files into decks, writing cards on-device with Apple Intelligence and re-syncing changed cards from linked Drive files.
- **BTC Survivor** (Python, NumPy) — always-on service that ingests live Coinbase order-book and trade streams, turns them into features every minute, and runs 20 learning bots, with checkpoint/resume, a replay harness, a read-only dashboard, and a systemd unit; runs 24/7 on a Linux server and passed a memory soak test (paper trading only).

ENGINEERING EXPERIENCE

Illini Electric Motorsports — *Firmware Developer, Battery Management System* Sep 2026 – Present
- Write real-time C firmware on FreeRTOS (built with Bazel) for the race car's battery management system: integrated the team's 1 Hz CAN heartbeat (bench-tested, merged) and developed a charging algorithm for the 140-cell pack, now in code review, that tapers current as the highest cell nears its voltage limit.

Lab AK — *Co-Founder and Software Engineer* Jan 2026 – Present
- Led development of Model mAKe Smith (labak.ai): ports, runs, and compares AI models on remote edge hardware through chip-vendor toolchains (Cadence NeuroWeave), including customers' own models; INT8 YOLOX-nano ran 25× faster with the Neo NPU (simulated).
- Led development of Model Analyzer (mma.labak.ai): inspects ONNX, TFLite, Keras, PyTorch, and TensorFlow models layer by layer, with weights and multiply-accumulate cost, entirely in the browser.

PilotCity — *Software Developer; Data Scientist* Nov 2022 – Aug 2024
- Fixed defects before deployment, optimized code paths, and cleaned and analyzed large datasets.

Seeed Studio — *Intern* Jun – Aug 2023
- Built edge-AI software for microprocessors, including an AI subject-tracking camera that kept tracking off-frame.

Alan AI — *Intern* Jul – Sep 2022
- Built a voice-activated GitHub FAQ on Alan AI's speech recognition platform, targeting a ~30% reduction in support call wait times.

RESEARCH EXPERIENCE

ASDRP — *Lead Researcher, Author, and Developer* Jun 2024 – Jan 2026
- Co-built ASDRP's iOS app (SwiftUI, Firebase; owned the student profile system) and built a Google Chat reminder bot that checks status-update quality with GPT.
- Co-first author of a paper under review at the *Journal of Emerging Investigators*; built the Python pipeline that processed 150 hours of audio across four speech recognition systems; at 1.25× speed (20% cheaper), Azure and Deepgram stayed under 0.10 WER.

Osteoporosis Gait Sensor — *Lead Developer and Designer* Jun 2024 – Present
- Designed and prototyped a wearable foot sensor that uses machine learning to detect early osteoporosis indicators from gait data.
- Built the instrument around a Raspberry Pi and a 16-sensor array laid out in the shape of a foot, with a real-time collection, preprocessing, and model-based analysis pipeline.

LEADERSHIP

Byte By Byte Coding — *CEO and Co-Founder* Jun 2024 – Present
- Co-founded a 501(c)(3) teaching computer science to 50+ underserved students; built the curriculum and lead volunteer instructors.

Boy Scouts of America — *Eagle Scout; Senior Patrol Leader; NYLT Staff* Aug 2018 – Jun 2026
- Taught leadership at National Youth Leadership Training; President's Volunteer Service Award (Gold) twice for 100+ service hours each.